PRACTICAL ASSIGNMENT REPORT

Cleaning and Analyzing Hotel Booking Data

Group: Nhóm 8 · Course: Data Analysis with Python · Lecturer: Dr. Tran Hung

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1. Objectives and Data Source

1.1. Objective

This report presents a descriptive analysis of booking records from two hotels in Portugal. It addresses ten questions on cancellations, seasonality of, booking lead time, guest country, market segments, deposit type, average daily rate (ADR), length of stay and room reassignment. The analysis is written from the viewpoint of a hotel group's revenue-management team. It describes and compares groups; it does not estimate causal effects.

1.2. Unit of analysis

One booking corresponds to one distinct booking_id. The raw file has 10,325 rows and 24 columns, but a row is an extraction record, not a booking: the file was assembled from several extractions, so one booking can appear in more than one row. After cleaning, the data contain 10,000 bookings: 6,702 (67.0%) at City Hotel and 3,298 (33.0%) at Resort Hotel.

1.3. Source and licence

The data come from Antonio, de Almeida and Nunes (2019), which documents booking demand at a resort hotel in the Algarve and a city hotel in Lisbon. The analysed file is a random sample of the 119,390 bookings in the original dataset (retrieved from the TidyTuesday repository, week of 11 February 2020). It was prepared for teaching: formatting errors, missing values, contradictory records and duplicate rows were inserted deliberately. The results therefore describe this sample and are not the hotels' actual operating performance.

Reference: Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. *Data in Brief*, 22, 41–49. https://doi.org/10.1016/j.dib.2018.11.126 (CC BY 4.0). Teaching version of the dataset, accessed 04/10/2026.

1.4. Period

Arrival dates run from 1 July 2015 to 31 August 2017; the latest reservation-status update in the source is 14 September 2017. Only 2016 covers all twelve months, so totals for 2015 and 2017 are not compared with 2016.

Variable definitions used in this report:

- **adr**: average daily rate, the booking's price per night in euro. It is not realised revenue.
- **is_canceled**: 1 for both canceled and no-show bookings.
- **lead_time**: number of days between the booking entry and the arrival date.

1.5. Limits of the data

The data contain no realised revenue or profit, no reasons for cancellation and no guest reviews. According to the source authors, guest country is usually confirmed at check-in, so it may be less reliable for cancelled bookings. The findings concern two hotels in one country during 2015–2017 and cannot be generalised to the hotel industry.

2. Data Quality and Processing

2.1 Overview before and after cleaning

Table 1. Data quality before and after cleaning.

Indicator	Before (n = 10,325 rows)	After (n = 10,000 bookings)	Note
Rows / bookings	10,325	10,000	Denominators differ (input rows vs. clean bookings).
Distinct booking_id	10,145	10,000	215 codes trimmed/uppercased (strip().upper()).
Duplicated on 23 cols	180 (raw) / 181 (norm)	0	1 duplicate hidden by case/whitespace issue.
Repeated booking_id	325	0	181 exact duplicates dropped (B1); 144 merged (B8).
City Hotel label	6,763	6,702	151 variant spellings normalized; deduplicated.
is_canceled contradicting reservation_status	41 rows	0	40 recomputed (B4); 1 dropped in B1.
Canceled bookings	3,769	3,741	Cleaned flags written as yes/True/1.0.
Cancellation rate	36.50%	37.41%	+0.91 pp change due to data cleaning.
Missing adr	0.24% (25 cells)	0.36% (36 cells)	11 values outside 0–1,000 or non-numeric set to missing.
Missing lead_time	0.10% (10 cells)	0.31% (31 cells)	21 values below 0 or above 800 days set to missing.
Missing country	1.23% (127 cells)	1.07% (107 cells)	113 of the 127 missing cells lie in representative rows; 6 of these were filled from another row of the same booking (B8).

Source: outputs/tables/q01_before_after.csv, outputs/quality_report.csv.

2.2 Three representative cleaning decisions

2.2.1 Decision 1. Defining the booking as the unit of analysis

Problem. booking_id had spaces or lower case in 215 cells (R00096, "BK044016 "), which inflated the number of distinct codes to 10,145. 181 rows repeated another row on all 23 business columns (R00608 repeats R00041), and 144 rows shared a booking_id with another row but differed in content. In 15 bookings the non-missing values disagreed: lead_time in 7 and adr in 8 (BK051775: adr 208.90 in R02322, 198.9 in R03865).

Rule. booking_id was trimmed and upper-cased before any comparison. Exact duplicates were removed, keeping the first row. The remaining rows with the same booking_id were merged column by column, taking the first non-missing value in ascending row_id order; this filled 22 missing cells. For the 15 conflicts, the value from the lowest row_id was kept and the conflict was flagged in the note column of row_lineage.csv.

Rationale. Duplicate detection on all 24 columns finds nothing because row_id is unique. The data dictionary treats bookings with different codes as different bookings, so rows were never merged on content alone: 1,759 bookings that match another booking on all other business columns were kept (a possible case is a group booking several identical rooms). The file does not say which extraction is authoritative, so a documented rule was used instead of averaging.

Impact. Without this step the data would hold 325 surplus rows (3.1% of the file), and duplicated records would count twice. Conflicts affect 15 of 10,000 bookings (0.15%).

2.2.2 Decision 2. Harmonising labels and reconciling is_canceled with reservation_status

Problem. After exact duplicates were removed, 865 rows (896 cells) carried a non-standard label in seven categorical columns (R00010, "city hotel"), and 272 cells in is_canceled and is_repeated_guest used yes/no, True/False or 1.0/0.0. In addition, 40 rows were contradictory: 29 with status Check-Out and 11 with Canceled or No-Show. In R00856, is_canceled is 1 while the status is Check-Out; the status date (20 September 2016) equals the arrival date (17 September 2016) plus the three booked nights.

Rule. Labels were mapped to the standard sets with dictionaries keyed on a normalised string; no unrecognised label remained (invalid count 0 in quality_report.csv). Where reservation_status was valid, is_canceled was recomputed from it (Check-Out → 0; Canceled and No-Show → 1); where it was missing, the standardised flag was kept. The meal label "Undefined" (101 rows) was recoded to SC, which the source documentation treats as the same thing.

Rationale. reservation_status is the final state recorded with a date, and the dates support it. Of the 29 contradictory Check-Out rows, 28 have a valid arrival date, status date and night count, and in all 28 the status date equals arrival plus booked nights. All 11 contradictory Canceled or No-Show rows have a status date on or before arrival. These rows were corrected, not deleted.

Impact. The reconciled rate is 37.41% (3,741 of 10,000 bookings), against 36.50% from the literal value "1" on raw rows. Exact label matching would also have missed 151 City Hotel rows with variant spellings.

2.2.3 Decision 3. Converting invalid dates and measurements to missing instead of correcting them

Problem. Arrival dates came in four formats: ISO (8,330 rows), dd/mm/yyyy (822), dd-Mon-yyyy (512) and "Month d, yyyy" (626); 25 were blank and 10 were not dates (R00030, "not_a_date"). Among the dates in the four formats, six were non-existent (R00169, "31/02/2016") and eight were outside the observation window (R00888, "2020-01-15"). adr carried a currency marker (€ or EUR) in 40 rows (R00081, "€114.30") and a decimal comma in 81 (R00432, "62,00"); 11 values were outside 0–1,000 or non-numeric (R00080, "-11"). A further 21 lead times were outside 0–800 days or non-numeric.

Rule. Each date format was parsed explicitly, reading dd/mm/yyyy day first; non-existent or out-of-window dates were set to missing with the reason logged. Currency markers were removed and the decimal comma replaced by a point (95,50 → 95.5); out-of-domain values were set to missing. Admissible but unusual values were kept: 132 bookings with ADR = 0 (120 non-canceled), lead times up to 709 days, 38 zero-night stays and 10 zero-guest bookings.

Rationale. The data support the day-first reading. Among 494 input rows with status Check-Out, a dd/mm/yyyy arrival date and a valid status date and night count, the status date minus the arrival date equals the booked nights

in all 494 rows under the day-first reading, but in only 52 (10.5%) under a month-first reading. Where no such evidence exists (R00888), the true value cannot be reconstructed.

Impact. The 49 bookings without a valid arrival date (0.49%) are excluded only from Question 3, and the 31 without lead time only from Question 4 (n = 9,969). ADR statistics (Question 8) rest on 6,121 non-canceled bookings with ADR > 0; 120 zero-ADR and 18 missing-ADR non-canceled bookings are reported separately.

2.3 Why missing values were retained

Missing values were kept as missing; they were not imputed and not used as grounds to delete records.

- **No evidence supports a replacement.** Assigning the 107 bookings with unknown country to the most common country (PRT) would raise Portugal's share from 40.05% to 41.12% with no supporting observation.
- **Deleting rows would discard usable information.** 347 bookings (3.47%) have at least one missing value in the business columns, yet most remain valid for analyses that do not use the missing field. Each analysis therefore uses all available cases for its variables and states its denominator (9,969 for lead time, 9,997 for market segment, 6,121 for ADR).
- **Missingness is low.** After cleaning, the highest rates among the original columns are 1.07% (country), 0.49% (arrival_date), 0.36% (adr) and 0.31% (lead_time) (Table A2).

Missing rates for adr, lead_time and arrival_date are higher after cleaning because values that were present but invalid are now recorded as missing. reservation_status is missing for 25 bookings because 26 input rows held only a date in that field; the date was moved to reservation_status_date (B2) and the status left empty. The label "Unknown" is used only when presenting results and is never stored. The process that generated the missing values was not tested, so an association between missingness and cancellation cannot be excluded.

3. Findings from the Data

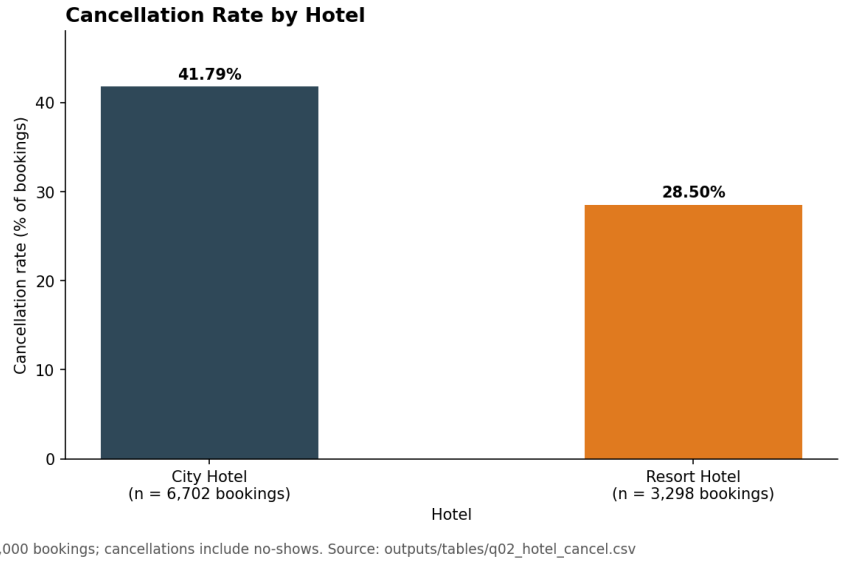
Six findings are reported, one per topic: data quality, hotel, seasonality, lead time, deposit type and market segment. Each states its denominator, a comparison and a limitation. All statements describe associations in a sample of two hotels in 2015–2017; none describes a cause.

3.1 Insight 1: Cleaning changed both the number of bookings and the measured cancellation rate

Cleaning changed both the booking count and the measured cancellation rate. The input contains 10,325 rows but only 10,000 distinct bookings: 181 exact duplicate rows were removed and 144 rows sharing a booking_id were merged, so 325 surplus rows (3.1% of the file) would otherwise have been counted twice (Table 1). A further 215 booking_id cells carried spaces or lower case, which inflated the number of distinct codes to 10,145 until strip().upper() was applied. The cancellation rate is 36.50% on raw rows (3,769/10,325, counting only the literal value “1”) and 37.41% on cleaned bookings (3,741/10,000), an increase of 0.91 percentage points (Table 2). Two corrections drive this shift: 105 cancellation flags written as yes/True/1.0 were standardized, and 40 flags that contradicted reservation_status were recomputed from it. The same issue affects hotel counts: exact-spelling matching finds only 6,763 City Hotel rows and misses 151 variant spellings. The missing rate of adr rose from 0.24% to 0.36% because 11 out-of-range values were set to missing rather than guessed. Within this two-hotel sample, every later rate therefore rests on the cleaned denominator of 10,000 bookings. Limitation: 15 bookings had conflicting lead_time (7) or adr (8) values across duplicate rows; the lowest-row_id rule is a documented convention, not proof of which value is correct.

3.2 Insight 2: City Hotel has a higher cancellation rate than Resort Hotel

Figure 1. Cancellation rate by hotel. n = 10,000 bookings



Source: outputs/tables/q02_hotel_cancel.csv.

Table 3. Bookings and cancellations by hotel.

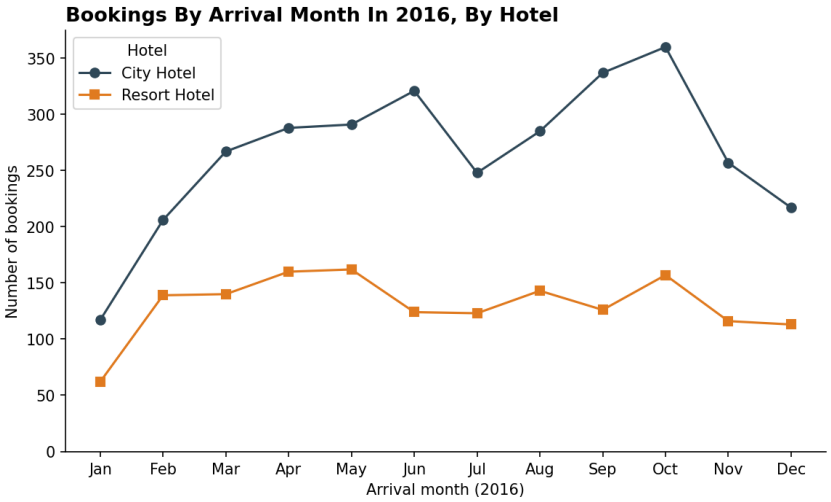
Hotel	Bookings	Share of bookings (%)	Canceled	Cancellation rate (%)
City Hotel	6,702	67.02	2,801	41.79
Resort Hotel	3,298	32.98	940	28.50

Source: outputs/tables/q02_hotel_cancel.csv.

City Hotel has a higher cancellation rate than Resort Hotel. Of the 10,000 cleaned bookings, 6,702 (67.02%) are City Hotel bookings, of which 2,801 were canceled (41.79%); Resort Hotel has 3,298 bookings, of which 940 were canceled (28.50%) (Table 3, Figure 1). The gap is 13.29 percentage points, and the City Hotel rate is 1.47 times the Resort Hotel rate. City Hotel accounts for 74.87% of all canceled bookings (2,801 of 3,741) against 67.02% of all bookings, so its weight in cancellations exceeds its weight in the sample by about 7.9 points. Both groups exceed 3,000 bookings, so the difference is not an artifact of a small subsample. The two hotels also differ in how far ahead guests book: median lead time is 74 days at City Hotel and 61 days at Resort Hotel (Table B2). Because lead time is associated with cancellation in this sample (Insight 4), that difference may partly overlap with the hotel gap, but the data cannot separate the two. Limitation: the hotels differ in type and location (Lisbon and the Algarve), and the data contain no cancellation reasons, deposit terms by hotel or channel mix by hotel, so the gap is described but not explained.

3.3 Insight 3: The 2016 monthly pattern differs between the two hotels

Figure 2. Bookings by arrival month in 2016, by hotel



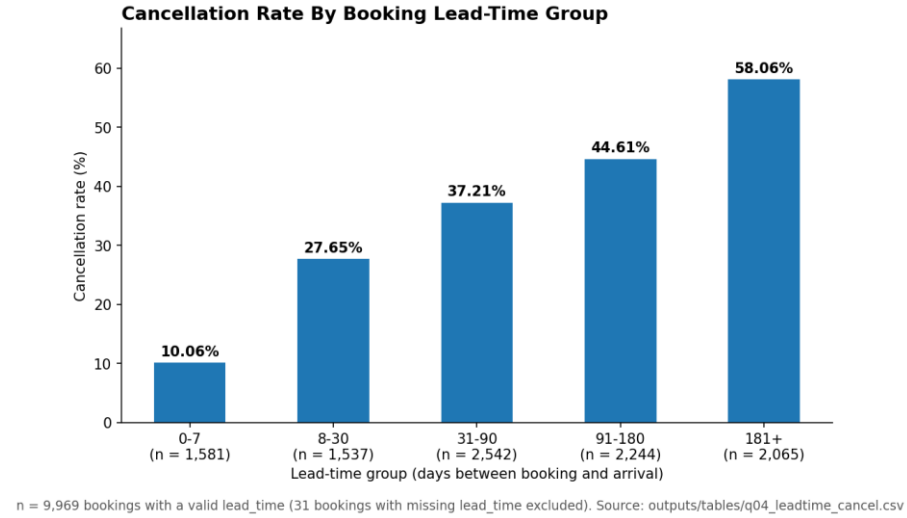
n = 4,759 bookings with a 2016 arrival date (canceled bookings included). Source: outputs/tables/q03_monthly_arrivals_2016.csv

Source: outputs/tables/q03_monthly_arrivals_2016.csv; monthly values in Table B1.

The 2016 monthly pattern differs between the two hotels. Of 4,759 bookings with a 2016 arrival date (canceled bookings included), City Hotel accounts for 3,194 and Resort Hotel for 1,565. City Hotel peaks in October with 360 bookings and is lowest in January with 117, a ratio of 3.08; Resort Hotel peaks in May with 162 and is lowest in January with 62, a ratio of 2.61 (Figure 2, Table B1). From February to December, Resort Hotel's monthly counts stay between 113 and 162, a narrow band, while City Hotel's range from 206 to 360, so City Hotel's volume rises more clearly in autumn. January is the weakest month for both hotels, with only 179 bookings combined. The combined cancellation rate also varies with the calendar: 40.62% in October (210 of 517 bookings), 40.17% in September and 39.29% in May, against 22.35% in January (40 of 179 bookings), a gap of 18.27 percentage points between October and January. July is a lower point (30.19%) despite being a summer month. Limitation: only one full year is available, so seasonality cannot be separated from year-specific events; the monthly counts include canceled bookings and are not guest-nights; and the monthly cancellation rate is not split by hotel.

3.4 Insight 4: Cancellation rates rise with booking lead time

Figure 3. Cancellation rate by booking lead-time group.



Source: outputs/tables/q04_leadtime_cancel.csv.

Table 4. Cancellation rate by lead-time group.

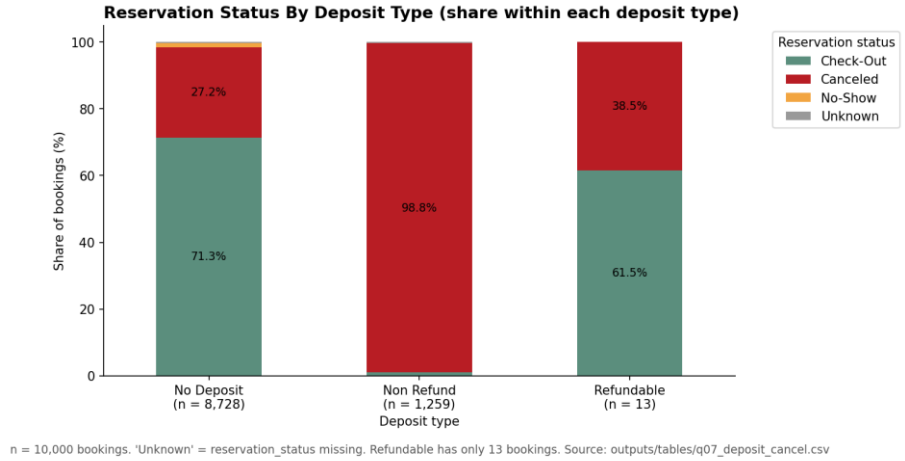
Lead-time group (days)	Bookings	Canceled	Cancellation rate (%)
0-7	1,581	159	10.06
8-30	1,537	425	27.65
31-90	2,542	946	37.21
91-180	2,244	1,001	44.61
181+	2,065	1,199	58.06

Source: outputs/tables/q04_leadtime_cancel.csv.

Cancellation rates rise with booking lead time. Across 9,969 bookings with a valid lead_time (31 excluded), the rate rises at every step across the five groups: 10.06% for 0-7 days (159 of 1,581 bookings), then 27.65% (425 of 1,537), 37.21% (946 of 2,542), 44.61% (1,001 of 2,244) and 58.06% for 181 days or more (1,199 of 2,065) (Table 4, Figure 3). The steps between adjacent groups are +17.59, +9.56, +7.40 and +13.45 percentage points, so the largest jump occurs between the shortest and second-shortest groups. The longest-lead group is 48.00 percentage points above the shortest, or 5.77 times its rate. The two longest groups together hold 4,309 bookings (43.2% of the 9,969), which means the high-lead range is a major part of the sample and not a marginal tail. Lead time also differs by hotel: the median is 74 days at City Hotel and 61 days at Resort Hotel, and the mean is 110.78 against 92.99 days (Table B2). Limitation: this is an observed association; it is not split by deposit type, market segment or hotel, and the data do not identify why guests cancel. The Non Refund and Groups results below suggest that some of this pattern may overlap with those variables.

3.5 Insight 5: Non Refund bookings are almost always recorded as canceled

Figure 4. Reservation status by deposit type (share within each deposit type)



Source: outputs/tables/q07_deposit_cancel.csv.

Table 5. Reservation status by deposit type.

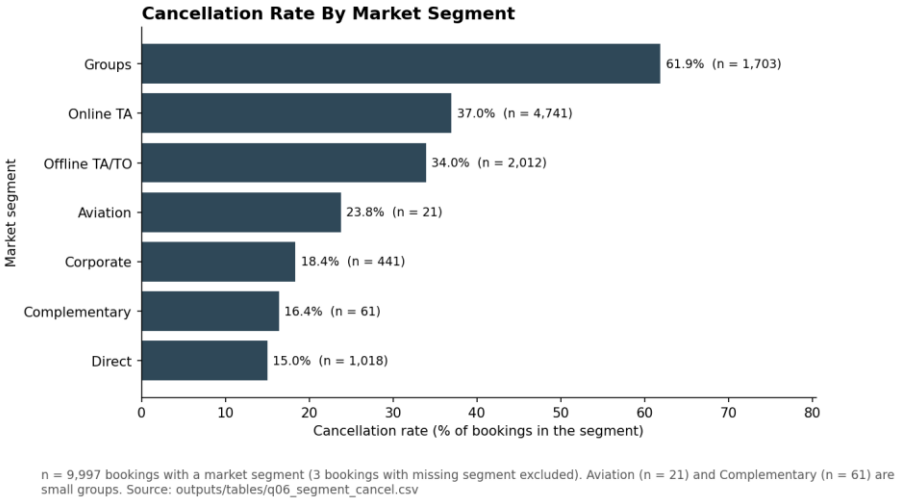
Deposit type	Bookings	Canceled, n (%)	Check-Out, n (%)	No-Show, n (%)	Status missing, n (%)
No Deposit	8,728	2,377 (27.23)	6,224 (71.31)	105 (1.20)	22 (0.25)
Non Refund	1,259	1,244 (98.81)	12 (0.95)	0 (0.00)	3 (0.24)
Refundable	13	5 (38.46)	8 (61.54)	0 (0.00)	0 (0.00)

Source: outputs/tables/q07_deposit_cancel.csv.

Non Refund bookings are almost always recorded as canceled. Of 1,259 Non Refund bookings, 1,244 (98.81%) have the status Canceled and 12 (0.95%) Check-Out; 3 have no status. Of 8,728 No Deposit bookings, 27.23% are Canceled (2,377), 71.31% Check-Out and 1.20% No-Show (105, all in this group) (Table 5, Figure 4). The gap in cancellation share between the two groups is 71.58 percentage points, or 3.63 times. Non Refund bookings make up 12.59% of all bookings but 34.31% of the 3,626 bookings with status Canceled, so a group of about one in eight bookings supplies about one in three cancellations. The group is also concentrated: of the 1,259 Non Refund bookings, 813 are Groups and 410 are Offline TA/TO, together 97.14% (Appendix D), so the deposit type is not spread evenly across the market segments. Refundable has only 13 bookings (5 Canceled, 8 Check-Out) and is too small for comparison. Limitation: the data do not allow the pattern to be attributed to the deposit itself. Deposit terms, refund rules and how a status is recorded for non-refundable rates are not documented in the file, and a status recorded automatically at booking could produce the same figures.

3.6 Insight 6: Groups has the highest cancellation rate, the gap narrows within one deposit type

Figure 5. Cancellation rate by market segment



Source: outputs/tables/q06_segment_cancel.csv.

Table 6. Bookings and cancellation rate by market segment (n = 9,997 bookings with a market segment; 3 without).

Market segment	Bookings	Share (%)	Canceled	Cancellation rate (%)
Online TA	4,741	47.42	1,753	36.98
Offline TA/TO	2,012	20.13	683	33.95
Groups	1,703	17.04	1,054	61.89
Direct	1,018	10.18	153	15.03
Corporate	441	4.41	81	18.37
Complementary	61	0.61	10	16.39
Aviation	21	0.21	5	23.81

Source: outputs/tables/q06_segment_cancel.csv.

Table 7. Cancellation rate within deposit type, five largest segments.

Market segment	No Deposit: bookings	No Deposit: rate (%)	Non Refund: bookings	Non Refund: rate (%)
Online TA	4,734	36.88	5 (n < 30)	100.00
Offline TA/TO	1,602	17.10	410	99.76
Groups	880	27.95	813	99.02
Direct	1,016	14.96	2 (n < 30)	50.00
Corporate	412	13.35	28 (n < 30)	92.86

Source: outputs/tables/x01_segment_deposit_cancel.csv (supplementary table)

Groups has the highest cancellation rate, but the gap to other segments narrows within one deposit type. Among 9,997 bookings with a market segment, Groups has 1,054 cancellations in 1,703 bookings (61.89%),

against 153 in 1,018 (15.03%) for Direct, a gap of 46.86 percentage points (Table 6, Figure 5). Groups is 17.04% of bookings but 28.19% of the 3,739 cancellations in this subset. However, 813 Groups bookings (47.74%) are Non Refund, with a cancellation rate of 99.02% (805 of 813). Within No Deposit, Groups cancels at 27.95% (246 of 880 bookings), 12.99 percentage points above Direct (14.96%, 1,016 bookings) and below Online TA (36.88%, 4,734 bookings) (Table 7). Offline TA/TO shows the same split: 17.10% under No Deposit (1,602 bookings) but 99.76% under Non Refund (410 bookings). Online TA, the largest segment at 47.42% of bookings, cancels 36.98% overall, close to its No Deposit rate because almost all of its bookings carry no deposit. Limitation: contract terms are not recorded, so the data cannot explain why Groups bookings concentrate in Non Refund; Aviation (n = 21) and Complementary (n = 61) are small, and most Non Refund cells outside Groups and Offline TA/TO have fewer than 30 bookings.

4. Limitations and Recommendations

4.1 Limitations

Figures depend on conventions and cleaning rules. No-Show is counted as canceled (105 bookings, all No Deposit); 40 contradictory flags were recomputed from reservation_status; 15 conflicting bookings were resolved by a lowest-row_id rule. Preparation alone moved the cancellation rate from 36.50% to 37.41%. Another valid rule set could give slightly different absolute values; comparisons between groups use the same rules throughout.

Seasonality rests on a single year. Only 2016 has all 12 months. Monthly counts include canceled bookings, are not numbers of guests staying, and are small in places (62 Resort Hotel bookings in January 2016).

Associations are observed, not isolated. Lead time, deposit type, market segment and country overlap, and no model separates them: 813 of 1,703 Groups bookings are Non Refund, and the Groups–Direct gap falls from 46.86 pp to 12.99 pp within No Deposit. The Portugal rate (57.68%, Appendix C) was not broken down by deposit type or segment. The data record no cancellation reasons.

adr and revenue_est are not revenue. adr is a nightly rate in EUR; revenue_est covers non-canceled bookings only and omits services, costs and refunds.

Quality of country. Guest country is usually confirmed at check-in, so it may be less accurate for canceled bookings; 107 bookings have no country. The code CN (101 input rows) was not merged with CHN, because the data give no evidence for doing so.

Small groups. Refundable has 13 bookings, Aviation 21 and Complementary 61; among Check-Out bookings with both room codes (Appendix G), room types B, C and H have 64, 50 and 23. Their rates are unstable.

Scope. The data cover two hotels in Portugal from July 2015 to August 2017 and are a sample into which errors were inserted deliberately; findings do not represent the hotel industry or forecast later years.

4.2 Recommendations for data and next questions

Cancellation reasons and timing. Record the reason for cancellation. The interval between cancellation and arrival can already be computed from reservation_status_date and arrival_date for canceled bookings.

Deposit and refund terms by segment and period. Obtain the conditions applied, to examine why Non Refund bookings are almost always recorded as canceled and why Groups bookings concentrate in Non Refund.

Several full years with an events calendar. At least two or three full years, with event calendars for Lisbon and the Algarve, would separate seasonality from year-to-year variation.

Further descriptive questions (no machine learning). (a) Cancellation rate by lead-time group × deposit type; (b) monthly cancellation rate by hotel, to check whether the 13.29 pp gap appears in most months; (c) bookings per segment × deposit-type cell, to flag unstable rates.

5. Division of Work

Member	Work completed	Parts the member can explain	Contribution
A: Ha Nhat Huy - 11253519	Data survey and missing/duplicate diagnostics (A1–A5); cleaning rules B1–B4; derived columns, monthly summary, and quality report (C1–C3); Questions 1–5 and Figures 1–4 (fig01–fig04); Insights 4–6; README.md draft.	All cleaning rules; Questions 1–5; at least two results from Questions 6–10.	50%
B: Le Minh Quan - 11257109	Issue catalog (A6); cleaning rules B5–B8 and row lineage tracking; automated validation and quality checks (C4–C4+); Questions 6–10 and Figures 5–8 (fig05–fig08); Insights 1–3; full report drafting, formatting, and final assembly.	All cleaning rules; Questions 6–10; at least two results from Questions 1–5.	50%

Each member reviewed the other's code, figures and text before the final run, and the notebook was re-run from the start with Restart & Run all.

Appendix

Appendix A. Cleaning log (20 logged rules)

Table A1. Cleaning log.

Step	Field(s)	Issue (summary)	Affected rows	Changed cells	Removed rows
B1	23 business columns	Exact duplicate rows	181	0	181
B2	reservation_status, reservation_status_date	Date written in the status column	26	52	0
B3	11 columns	Empty, missing-value codes and "Undefined" (segment, channel) set to missing	286	289	0
B4	7 categorical columns	Non-standard labels	865	896	0
B4	meal	"Undefined" recoded to SC	101	101	0
B4	is_canceled, is_repeated_guest	yes/no, True/False, 1.0/0.0 recoded to 0/1	272	272	0
B4	reserved_room_type, assigned_room_type	Lower case, extra spaces, codes outside A–P	114	114	0
B4	is_canceled	Contradiction with reservation_status	40	40	0
B5	arrival_date	Invalid or outside 2015-07-01 to 2017-08-31	49	24	0
B5	reservation_status_date	Invalid or outside 2014-10-01 to 2017-09-14	10	10	0
B6	lead_time	Outside 0–800 or non-numeric	21	21	0
B6	stays_in_weekend_nights	Outside 0–20 or non-numeric	10	10	0
B6	stays_in_week_nights	Outside 0–50 or non-numeric	7	7	0
B6	adults	Outside 0–10 or non-numeric	0	0	0
B6	children	Outside 0–10 or non-numeric	0	0	0
B6	babies	Outside 0–10 or non-numeric	5	5	0
B6	total_of_special_requests	Outside 0–5 or non-numeric	0	0	0
B6	adr	Currency symbol, decimal comma, outside 0–1,000	11	11	0
B7	country	Non-standard code, full name or missing-value code	199	199	0
B8	booking_id	Rows with the same booking merged (288 rows involved)	288	22	144

Source: outputs/cleaning_log.csv. A row can meet several detection rules, so affected_rows must not be summed across rules. For B5 arrival_date, 49 rows meet the rule, of which 25 were already blank, so 24 cells changed value.

Table A2. Missing values before and after cleaning (selected columns).

Column	Before: missing (n = 10,325 rows)	Before: invalid values set to missing	After: missing (n = 10,000 bookings)
country	127 (1.23%)	0	107 (1.07%)
arrival_date	25 (0.24%)	24	49 (0.49%)
adr	25 (0.24%)	11	36 (0.36%)
lead_time	10 (0.10%)	21	31 (0.31%)
reservation_status	0 (0.00%)	26 (dates in the wrong column)	25 (0.25%)

Source: outputs/quality_report.csv. "Invalid" means a value that was present but violated the data conventions.

Appendix B. Supplementary results for Questions 3 and 4

Table B1. Bookings by arrival month, 2016 (Question 3). *Denominator: 4,759 bookings with a 2016 arrival date; the cancellation rate is computed on all bookings of the month, both hotels combined.*

Month	City Hotel	Resort Hotel	Total	Cancellation rate (%)
1	117	62	179	22.35
2	206	139	345	33.04
3	267	140	407	33.42
4	288	160	448	37.28
5	291	162	453	39.29
6	321	124	445	37.30
7	248	123	371	30.19
8	285	143	428	36.68
9	337	126	463	40.17
10	360	157	517	40.62
11	257	116	373	39.68
12	217	113	330	37.58
Total	**3,194**	**1,565**	**4,759**	

Source: outputs/tables/q03_monthly_arrivals_2016.csv.

Table B2. Lead time (days) by hotel (Question 4). *Denominator: bookings with a valid lead_time (n = 9,969).*

Hotel	Bookings with valid lead time	Mean	Median	Q1	Q3	Max
City Hotel	6,683	110.78	74	24	164	629
Resort Hotel	3,286	92.99	61	12	154	709

Source: outputs/tables/q04_leadtime_stats.csv.

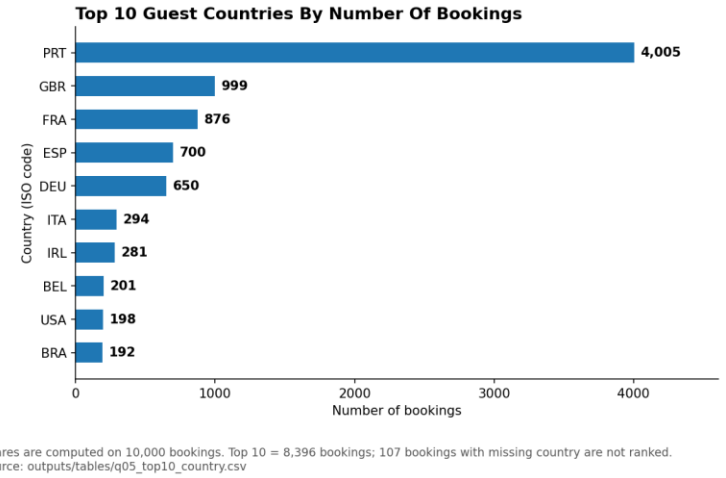
Appendix C. Top 10 guest countries (Question 5)

Table C1. Top 10 countries by number of bookings.

Rank	Country	Bookings	Share of all 10,000 bookings (%)	Cancellation rate (%)
1	PRT	4,005	40.05	57.68
2	GBR	999	9.99	19.72
3	FRA	876	8.76	19.06
4	ESP	700	7.00	24.57
5	DEU	650	6.50	18.62
6	ITA	294	2.94	33.67
7	IRL	281	2.81	23.84
8	BEL	201	2.01	18.41
9	USA	198	1.98	24.75
10	BRA	192	1.92	41.15

Source: outputs/tables/q05_top10_country.csv.

Figure C1. *Top 10 guest countries by number of bookings*



Source: outputs/tables/q05_top10_country.csv.

9,893 bookings have a country code and 107 do not (reported separately). The Top 10 holds 8,396 bookings (83.96% of all bookings). Portugal (PRT) is the largest market, with 4,005 bookings (40.05%) and a cancellation rate of 57.68%; PRT accounts for 2,310 of the 3,741 canceled bookings (61.75%). GBR, FRA, ESP and DEU have cancellation rates between 18.62% and 24.57%. The data contain no information on the composition of guests within a country, and the Portugal rate was not split by deposit type or segment.

Appendix D. Cancellation rate by market segment and deposit type (Question 6, supplementary)

Table D1. Cancellation rate by market segment and deposit type. *Denominator: bookings in each segment × deposit-type cell.*

Market segment	Deposit type	Bookings	Canceled	Cancellation rate (%)
Aviation	No Deposit	21	5	23.81
Complementary	No Deposit	61	10	16.39

Market segment	Deposit type	Bookings	Canceled	Cancellation rate (%)
Corporate	No Deposit	412	55	13.35
Corporate	Non Refund	28	26	92.86
Corporate	Refundable	1	0	0.00
Direct	No Deposit	1,016	152	14.96
Direct	Non Refund	2	1	50.00
Groups	No Deposit	880	246	27.95
Groups	Non Refund	813	805	99.02
Groups	Refundable	10	3	30.00
Offline TA/TO	No Deposit	1,602	274	17.10
Offline TA/TO	Non Refund	410	409	99.76
Online TA	No Deposit	4,734	1,746	36.88
Online TA	Non Refund	5	5	100.00
Online TA	Refundable	2	2	100.00

Source: outputs/tables/x01_segment_deposit_cancel.csv. Cells with fewer than 30 bookings are unstable.

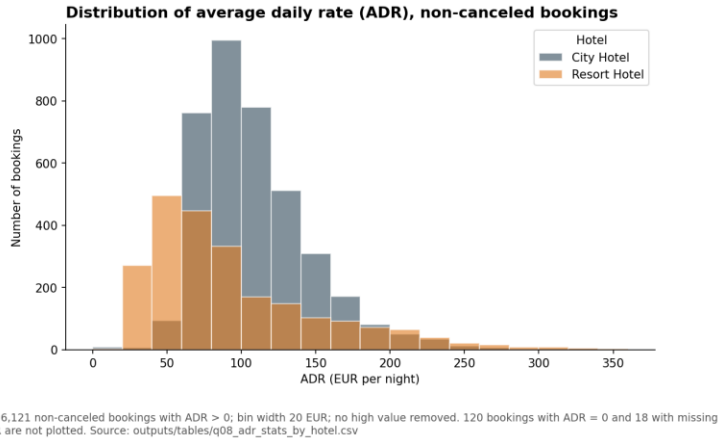
Appendix E. Average daily rate, non-canceled bookings (Question 8)

Table E1. ADR statistics by hotel (EUR per night).

Hotel	Bookings	Mean	Median	Min	Max	Q1	Q3
City Hotel	3,828	108.18	100.65	4.50	336.00	80.75	126.90
Resort Hotel	2,293	93.60	75.00	6.00	349.67	51.50	120.00

Source: outputs/tables/q08_adr_stats_by_hotel.csv.

Figure E1. *Distribution of average daily rate (ADR), non-canceled bookings*



Source: outputs/tables/q08_adr_stats_by_hotel.csv.

Of 6,259 non-canceled bookings, 6,121 have ADR > 0, 120 have ADR = 0 and 18 have a missing ADR. The central half of ADR values (Q1–Q3) spans 51.50–120.00 EUR at Resort Hotel (width 68.50) and 80.75–126.90 EUR at City Hotel (width 46.15). The mean exceeds the median at both hotels (City Hotel by 7.53 EUR, Resort

Hotel by 18.60 EUR), which indicates a right-skewed distribution, more strongly at Resort Hotel. ADR is a nightly rate, not revenue, and the data are not split by season.

Appendix F. Length of stay, non-canceled bookings (Question 9)

Table F1. Share of stays of 1–3 nights.

Hotel	Non-canceled bookings with valid total_nights	Stays of 1–3 nights	Share (%)
City Hotel	3,884	2,730	70.29
Resort Hotel	2,350	1,192	50.72
Both hotels	6,234	3,922	62.91

Source: outputs/tables/q09_stay_nights.csv.

Figure F1. *Distribution of booked nights, non-canceled bookings*



Source: outputs/tables/q09_stay_nights.csv.

Stays of 1–3 nights are the majority at both hotels, more clearly at City Hotel. Resort Hotel shows a separate cluster at 7 nights: 406 bookings (17.28% of 2,350), more than the 4-night (249) and 5-night (153) groups of the same hotel, whereas City Hotel has 107 seven-night bookings (2.75% of 3,884). The data contain no trip purpose, so the 7-night cluster is not explained. Booked nights are the nights reserved, not necessarily the nights stayed.

Appendix G. Room type changes at check-out (Question 10)

Table G1. Room type change rate by hotel. *Denominator: Check-Out bookings with both room codes valid (n = 6,242).*

Hotel	Bookings	Changed	Change rate (%)
City Hotel	3,891	561	14.42
Resort Hotel	2,351	583	24.80

Table G2. Room type change rate by reserved room type.

Reserved room type	Bookings	Changed	Change rate (%)
A	4,287	948	22.11
D	1,137	114	10.03
E	397	47	11.84

Reserved room type	Bookings	Changed	Change rate (%)
F	174	15	8.62
G	110	3	2.73
B	64	11	17.19
C	50	4	8.00
H	23	2	8.70

Source: outputs/tables/q10_room_change_by_hotel.csv, outputs/tables/q10_room_change_by_reserved_type.csv.

The change rate is 24.80% at Resort Hotel and 14.42% at City Hotel, a gap of 10.38 pp; over both hotels it is 1,144 of 6,242 (18.33%). Room type A is the largest group (4,287 bookings) and has the highest change rate (22.11%) among the types with at least 100 bookings. Types B, C and H have fewer than 70 bookings, so their rates are unstable. The data record only whether the room type changed, not the reason (upgrade, downgrade, lack of availability or guest request). The denominator of this question (6,242) differs from that of Question 9 (6,234) because the populations differ: Question 10 uses 6,242 Check-Out bookings that have two valid room codes, whereas Question 9 uses 6,234 non-canceled bookings that have a valid number of nights. The broader non-canceled population contains 6,259 bookings; 25 have missing total_nights and are excluded from Question 9.